Lecture 3. Gains and Losses from Trade in the Specific-Factors Model

Lectures on International Trade

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Recall from Lecture 2

Gains from trade in the Ricardian model:

- Every country gains from trade.
- Every worker gains from trade.
- In reality, we see trade generates winners and losers.
- Problem: only one factor in the Ricardian model, i.e. labor
- In order to analyse the gains and losses in trade, we need more production factors.
  - The Specific-Factors Model (Lecture 3)
  - The Heckscher-Ohlin Model (Lecture 4)

Outline

Short-run analysis of trade effects:

- The Specific-Factors Model
  - Model settings
  - Equilibrium of factor allocation
  - Factor returns
- Effects of trade on factor returns

Motivating Facts

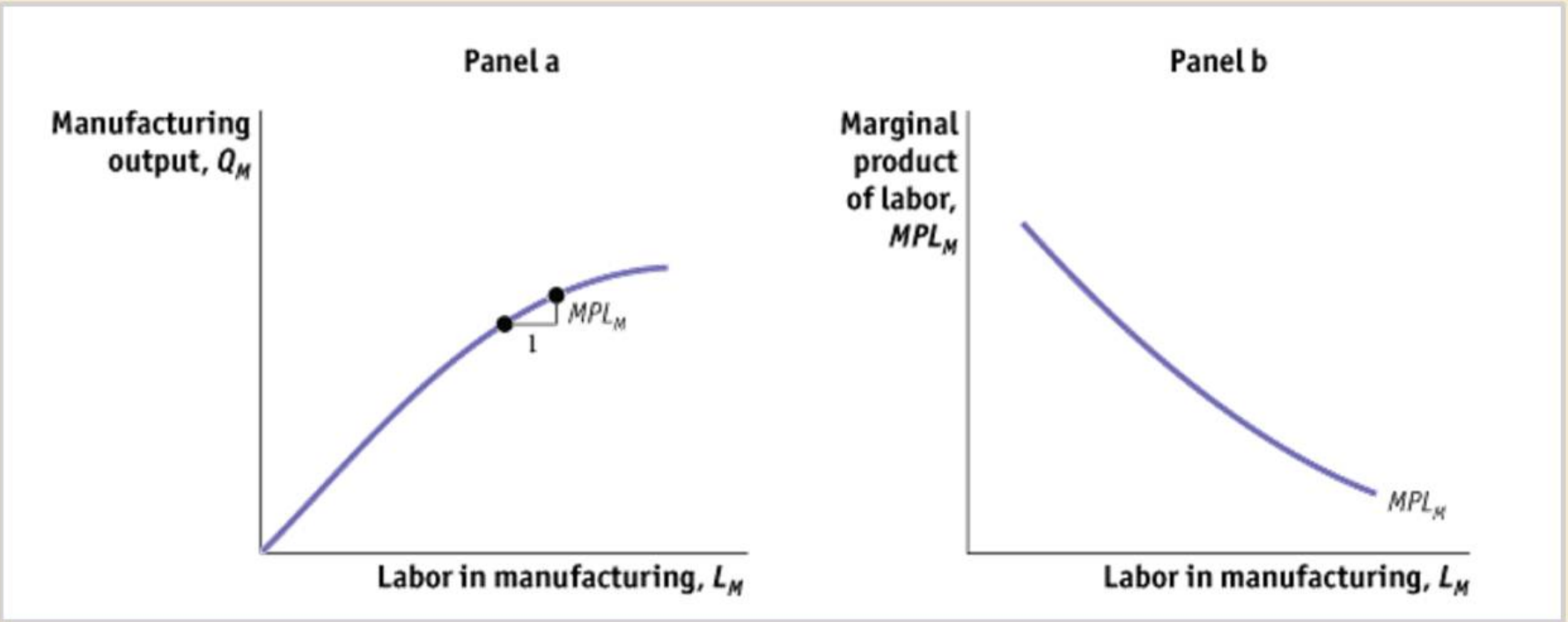
- Many factors in production.
- Some factors are mobile, some are not, at least not perfectly.
  - Mobile: labor
  - Less mobile: e.g. land to agriculture, equipment to manufacturing.
- Trade increase the price of good that one is exporting (comparative advantage good).
- Return to factors producing that good will sure increase.

The Specific-Factors Model

- Factors of production
  - Specific-factors: Land in Agriculture; Capital in Manufacturing
  - Mobile factor: Labor
  - Agriculture industry uses labor and land; Manufacturing uses labor and capital.
- Two countries: Home and Foreign
- If endowments of capital and land are constant, the marginal product of labor declines as the amount of labor used in the industry increases.

Decreasing MPL

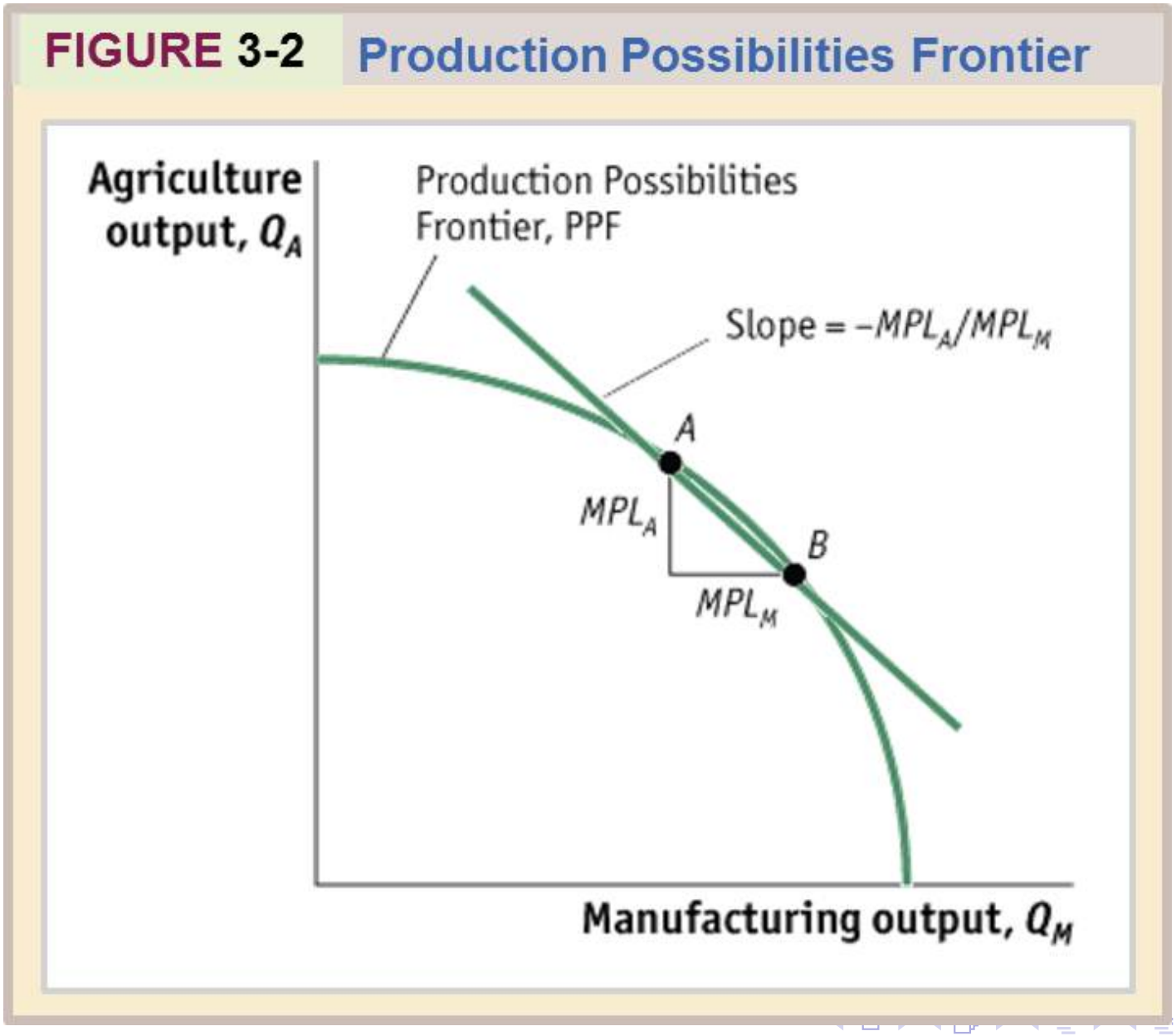
FIGURE 3-1



Panel (a) Manufacturing Output As more labor is used, manufacturing output increases, but it does so at a diminishing rate.

Panel (b) Diminishing Marginal Product of Labor An increase in the amount of labor used in manufacturing lowers the marginal product of labor.

PPF of Home

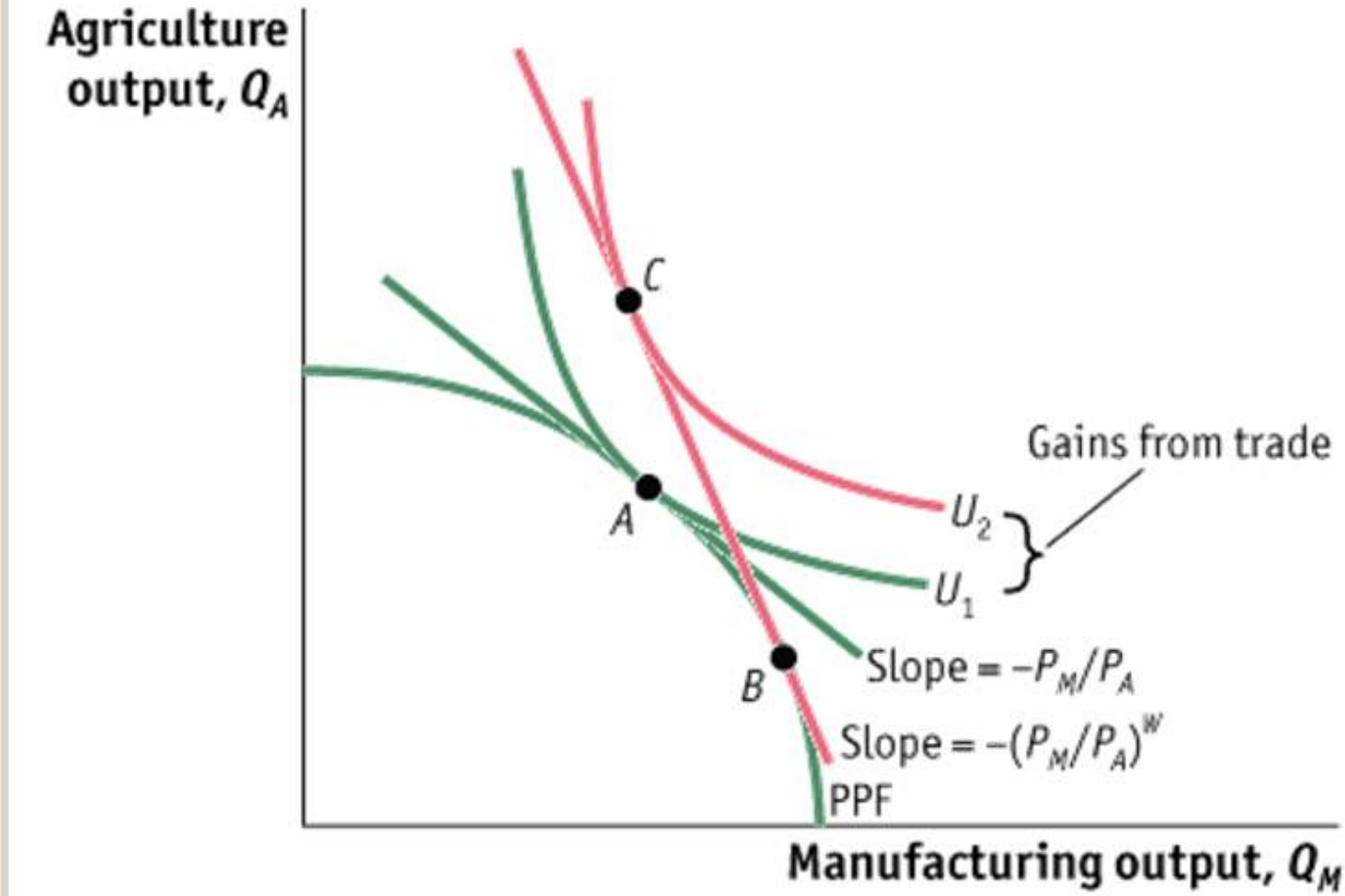

FIGURE 3-2
Production Possibilities Frontier
Agriculture output, Q_A
Production Possibilities Frontier, PPF
Slope = $-MPL_A/MPL_M$
A
B
MPL_A
MPL_M
Manufacturing output, Q_M

The Specific-Factors Model, cont'

- As in the Ricardian model, the slope of the PPF equals the opportunity cost or relative price of the good on the horizontal axis: here it is manufacturing output.
- Firms hire labor up to the point where the cost of one more hour of labor (the wage) equals the value of one more hour of labor in production.
- $W = P_M \cdot MPL_M$ and $W = P_A \cdot MPL_A$
- Assume that the Home no-trade relative price of manufacturing is lower than the Foreign relative price, $(P_M/P_A) < (P_M^*/P_A^*)$.
- Thus Home has comparative advantage in manufacturing.
- With trade, the relative price of exported good (manufacturing, for Home) increases; The relative price of imported good (agriculture, for Home) decreases.

Home Economy with Trade

FIGURE 3-3 Home has Comparative Advantage in M



The Specific-Factors Model, cont'

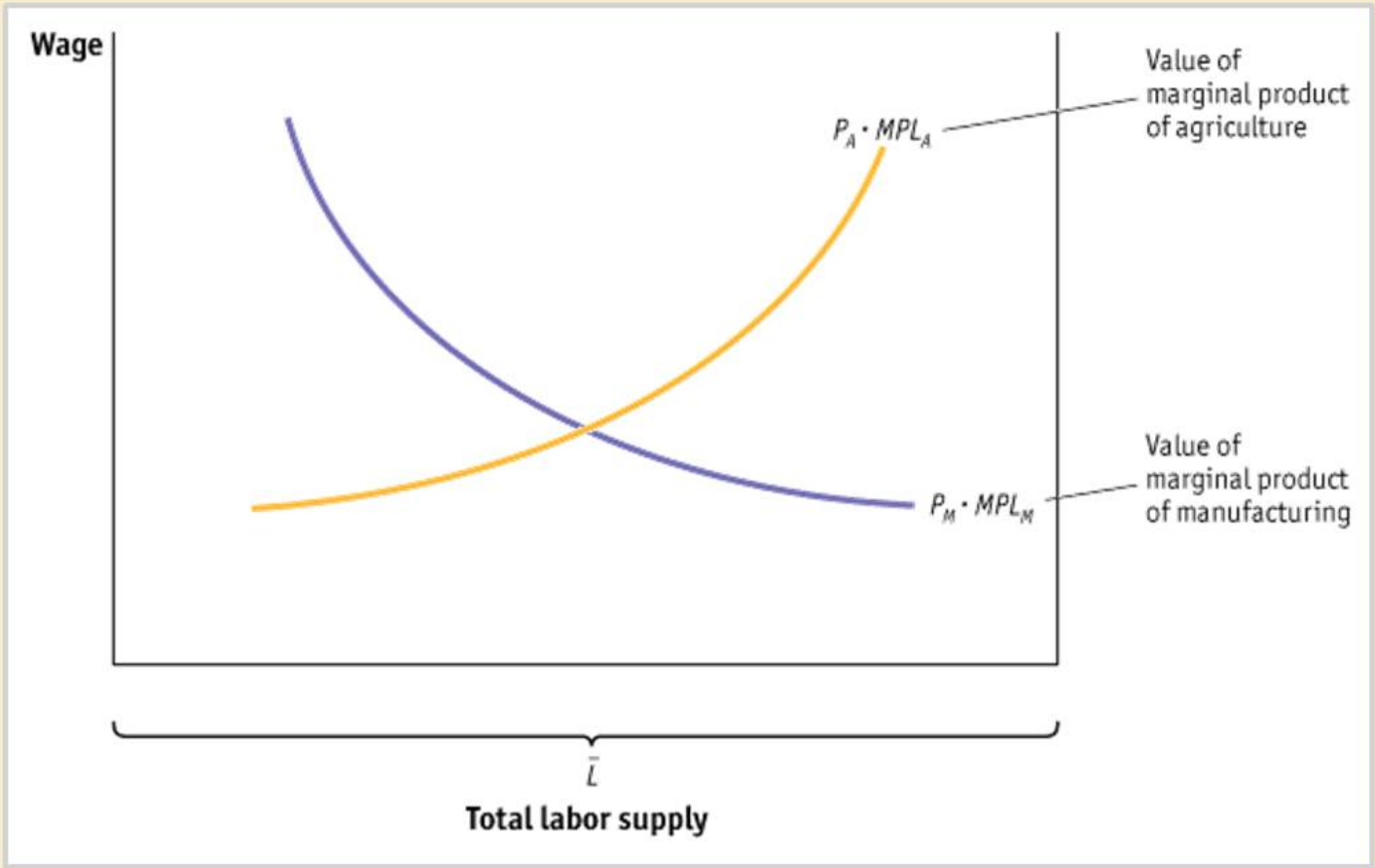
Determination of wages

- Wages are equal across two industries.
- $W = P_M \cdot MPL_M = P_A \cdot MPL_A$

Determination of Wages

A useful figure to show labor allocation:

FIGURE 3-4 (1 of 2) **Allocation of Labor between Manufacturing and Agriculture**

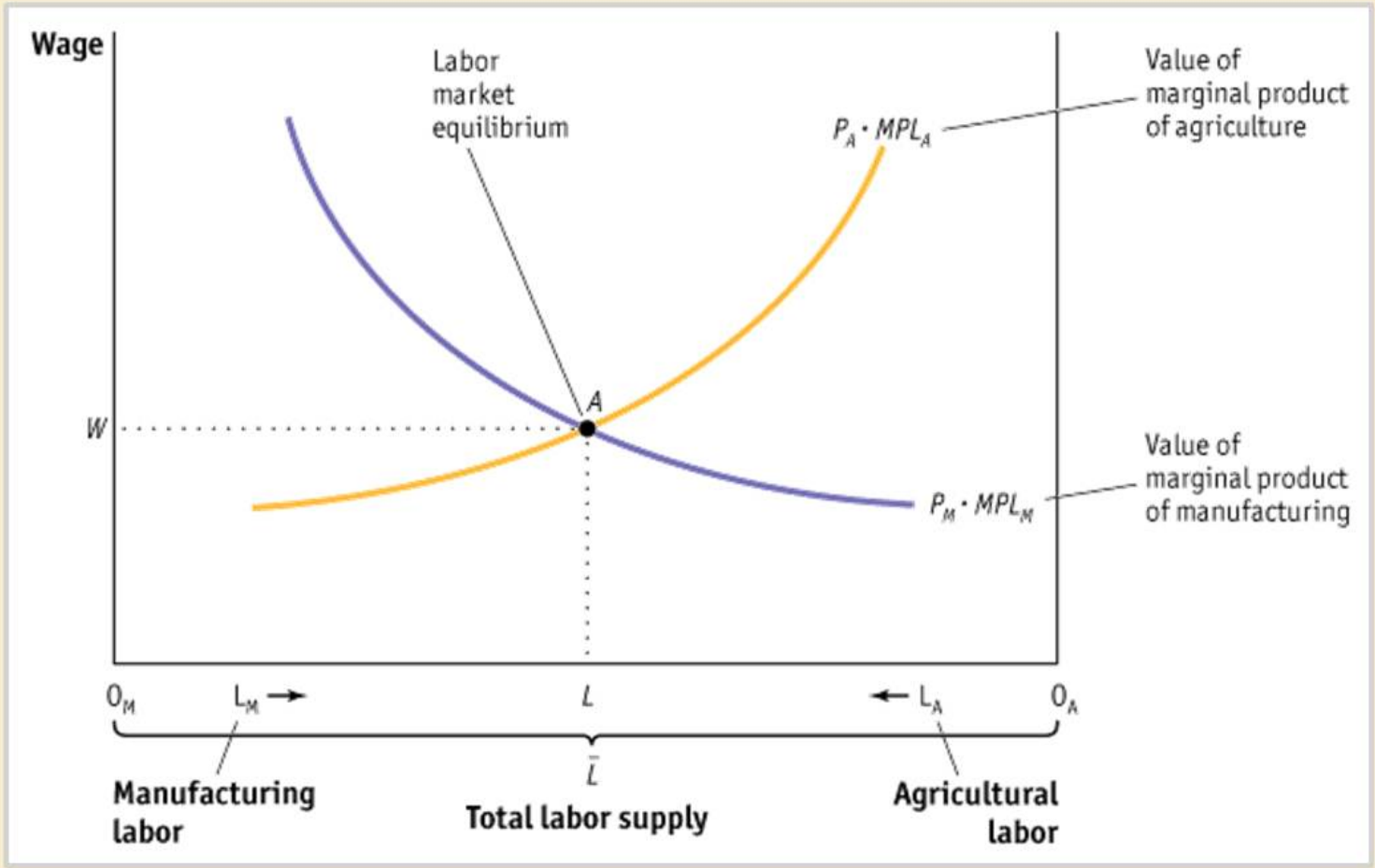


The amount of labor used in manufacturing is measured from left to right along the horizontal axis, and the amount of labor used in agriculture is measured from right to left.

Determination of Wages, cont'

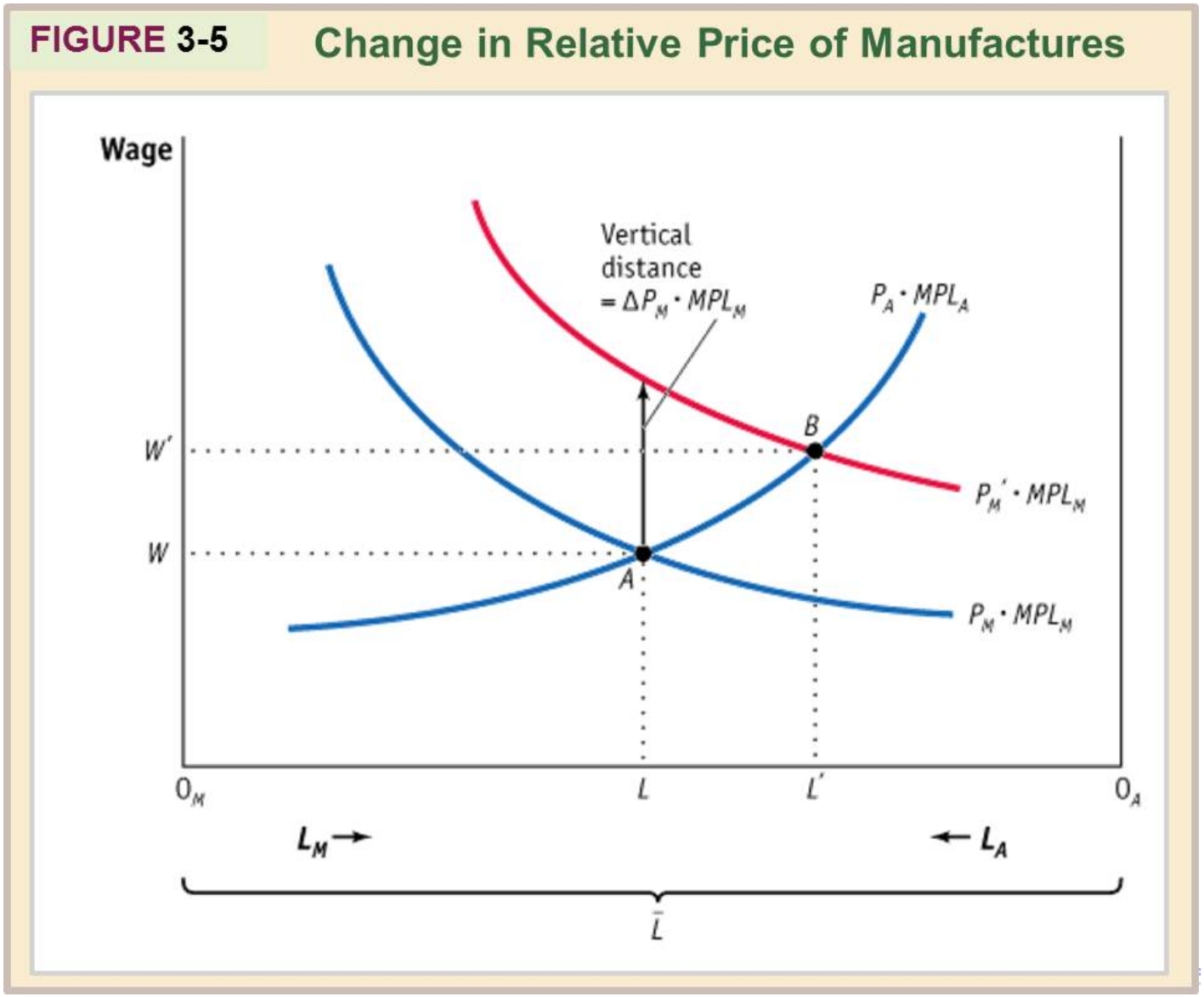
Labor market equilibrium:

FIGURE 3-4 (2 of 2) Allocation of Labor between Manufacturing and Agriculture (continued)



Determination of Wages, with Trade

Labor market equilibrium with trade:



Earnings of Labor after Trade

- Wage level increases from W to W'.
- But this increase is less than the upward shift $\Delta P_M \cdot MPL_M$.
- In the case shown in Figure 3-5, P_M increase, P_A stays the same.

Earnings of Labor after Trade

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- So real wage in terms of food W/P_A increases to W'/P_A.
- How about real wage in terms of manufactures?

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  - $W/P_M > W'/P'_M$
- Overall impact on real wage: *ambiguous*
- This is different from the Ricardian model.
- This warns us that one cannot make unqualified statements about the effects of trade on workers. The effect of trade on real wages can be complex.

The Specific-Factors Model, cont'

Other labor market outcome:

- Specific-factors model is also related with unemployment.
- Structural change: decreasing share of manufacturing, increasing share of service.
- Time to find new job for relocated labor.
- Job-loss due to import competition: **TAA** Trade Adjustment Assistance.

Returns to Specific Factors

- As more labor is used in manufacturing, the marginal product of capital will rise because each machine has more labor to work it.
- As labor leaves agriculture, the marginal product of land will fall because each acre of land has fewer laborers to work it.
- An increase in the quantity of labor used in an industry will raise the marginal product of the factor specific to that industry, and a decrease in labor will lower the marginal product of the specific factor.
- Real rental for capital in terms of manufactured good: $P_M \cdot MPK_M/P_M = MPK_M$ increases.
- Real rental for land in terms of food: $MPLand_A$ decreases.

Returns to Specific Factors, cont'

- An increase in the relative price of an industry's output will increase the real rental earned by the factor specific to that industry but will decrease the real rental of factors specific to other industries.

Returns to Specific Factors, cont'

- An increase in the relative price of an industry's output will increase the real rental earned by the factor specific to that industry but will decrease the real rental of factors specific to other industries.
- This conclusion means that:
  - The specific factors used in export industries will generally gain as trade is opened, since export good price rises.
  - The specific factors used in import industries will generally lose as trade is opened, since import good price falls.

Summary

Key Points:

1. Opening a country to international trade leads to overall gains, but in a model with several factors of production, some factors of production will lose.

2. The fact that some people are harmed because of trade sometimes creates social tensions that may be strong enough to topple governments. A recent example is Bolivia, where the citizens cannot agree on how to share the gains from exporting natural gas.

Summary, cont'

3. In the specific-factors model, factors of production that cannot move between industries will gain or lose the most from opening a country to trade. The factor of production that is specific to the import industry will lose in real terms, as the relative price of the import good falls. The factor of production that is specific to the export industry will gain in real terms, as the relative price of the export good rises.

4. In the specific-factors model, labor can move between the industries and earns the same wage in each. When the relative price of either good changes, then the real wage rises when measured in terms of one good but falls when measured in terms of the other good. Without knowing how much of each good workers prefer to consume, we cannot say whether workers are better off or worse off because of trade.

Summary, cont'

5. Economists do not normally count the costs of unemployment as a loss from trade because people are often able to find new jobs. In the United States, for example, about two-thirds of people laid off from manufacturing or services find new jobs within two or three years, though sometimes at lower wages.

6. Trade Adjustment Assistance policies are intended to compensate those who are harmed due to trade by providing additional income during the period of unemployment. Recently, the Trade Adjustment Assistance program in the United States was expanded to include workers laid off due to trade in service industries.

7. Even when many people are employed in export activities, such as those involved in coffee export from certain developing countries, fluctuations in the world market price can lead to large changes in income for growers and workers.